Probabilistic First-Order Logic (PFOL) Specification

Matthew Andrews

December 13, 2025

Abstract

This document provides a unified mathematical specification of the MEGA(Mutable-Encoding Genetic Algorithm) framework. It describes the system as implemented in the canonical code base, integrating hash encoding, dynamic meta gene formation, delimiter-aware operators, and LRU-based pattern management. All specifications match the reference implementation exactly.

Contents

1 Parameter Reference **3**

2 The Signature: Language of the Theory ($\mathcal{L}$) **3**
- 2.1 Sorts (Domains) . . . 3
- 2.2 Functions . . . 3
- 2.3 Predicates . . . 4
- 2.4 Delimiter Pairing Constraint . . . 4

3 Structural Axioms **4**
- 3.1 A0. Initial Population Constraint . . . 4
- 3.2 A1. Search Space Geometry . . . 5
- 3.3 A2. Hash Encoding System . . . 5
- 3.4 A3. Meta gene State Management . . . 6

4 Probabilistic Axioms **6**
- 4.1 P1. Crossover . . . 7
- 4.2 P2. Mutation . . . 7
- 4.3 P3. Gene Selection . . . 8

5 Metagene Operations **8**
- 5.1 A4. Meta gene Capture . . . 8
- 5.2 A5. Meta gene Open . . . 9

6 Population Structure **9**
- 6.1 A6. Elitism and Special Organisms . . . 9

7 Edge Cases and Fallback Behaviors **10**
- 7.1 A7. Edge Case Specifications . . . 10

8 Initialization Process **10**
8.1 A8. Population Initialization 10

9 System Evolution **10**
9.1 A9. Total System Evolution (Ψ) 10

10 Geometric Interpretation **12**
10.1 A10. Hyperbolic Model 12

11 Conservation Laws **12**
11.1 A11. Invariants 12

12 Summary **13**

1 Parameter Reference

Table 1 defines the complete parameter space $\Pi = \{\pi_1, \ldots, \pi_{15}\}$.

ID	Code Variable	Type	Semantic Meaning
π_1	`population_size`	int	Population size
π_2	`num_parents`	int	Number of parents selected
π_3	`max_individual_length`	int	Maximum organism length
π_4	`mutation_prob`	float	Base mutation probability
π_5	`delimited_mutation_prob`	float	Delimited region mutation prob
π_6	`delimit_delete_prob`	float	Delimiter deletion probability
π_7	`open_mutation_prob`	float	Meta gene open probability
π_8	`metagene_mutation_prob`	float	Capture probability
π_9	`delimiter_insert_prob`	float	Delimiter insertion probability
π_{10}	`crossover_prob`	float	Crossover probability
π_{11}	`elitism_ratio`	float	Elite fraction preserved
π_{12}	`base_gene_prob`	float	Base gene selection weight
π_{13}	`metagene_prob`	float	Meta gene exponential weight
π_{14}	`lru_cache_size`	int	Active set capacity
π_{15}	(hardcoded: 2)	int	Deletion basket age threshold

Table 1: MEGA parameter definitions

2 The Signature: Language of the Theory ($\mathcal{L}$)

2.1 Sorts (Domains)

- $\mathcal{T}$: Time (generations $\mathbb{N}$)
- $\mathcal{E}$: Encodings (64-bit hash keys)
- $\mathcal{E}'$: Padded encodings ($\mathcal{E} \cup \{0\}$)
- $\mathcal{X}$: Organisms (finite sequences over $\mathcal{E}$)
- Π: Parameters (fixed vector)
- $\mathcal{G}$: Generation counter

2.2 Functions

- $\text{len}(x)$: Length of organism x
- $\xi(e)$: Delimiter-free constituent sequence of meta gene e
- $\beta(x, j)$: Delimiter depth at position j in x
- $D_{\max}(t)$: Maximum organism length observed by time t
- $L(e, t)$: Last access time of encoding e

- $\text{Age}_{\mathcal{T}}(e, t)$: Generations e has been in deletion basket
- $H(i)$: Hash function $\mathbb{N} \to \mathbb{Z}_{64}$
- $\text{Decode}(x)$: Fully expanded, delimiter-free sequence
- $F(x)$: Fitness function
- $\text{DepthZeroCuts}(x)$: Valid crossover positions in x
- $\text{Deduplicate}(\sigma, \mathcal{M})$: Existing meta gene with content σ, or $\perp$
- $S(P_t)$: Deterministic parent selection (top π_2 by fitness)
- $\text{Pair}(i, S)$: Deterministic pairing function for i-th offspring

2.3 Predicates

- $\text{InPop}(x, t)$: Organism x is in population P_t
- $\text{StateA}(e, t)$: Encoding e is in Active set $\mathcal{A}(t)$
- $\text{StateT}(e, t)$: Encoding e is in deletion basket $\mathcal{T}(t)$
- $\text{StateP}(e, t)$: Encoding e is in reuse pool $\mathcal{P}(t)$
- $\text{IsMetageneEncoding}(e, t)$: e is a metagene at time t
- $\text{IsElite}(x, t)$: Organism x is in elite set at time t
- $\text{IsFullyDelimited}(x)$: x starts with Start, ends with End
- $\text{NonDelimitedCut}(x, j)$: Position j is valid for crossover

Formal definitions:

$$\text{LeastRecentlyUsed}(e, t) \Leftrightarrow \text{StateA}(e, t) \wedge \forall e' \in \mathcal{A}(t).L(e, t) \leq L(e', t) \tag{1}$$

$$\text{NonDelimitedCut}(x, j) \Leftrightarrow 1 \leq j \leq \text{len}(x) \wedge \beta(x, j-1) = 0 \tag{2}$$

2.4 Delimiter Pairing Constraint

Delimiters (Start, End) are treated as atomic matched pairs. All genetic operations preserve structural pairing and directionality. Substituting Start for End (or vice versa) is forbidden.

3 Structural Axioms

3.1 A0. Initial Population Constraint

$$\forall x \, [\text{InPop}(x, 0) \to (2 \leq \text{len}(x) \leq \pi_3 \wedge \text{Valid}(x))] \tag{3}$$

This constraint applies only at $t = 0$. No upper bound is imposed for later generations.

3.2 A1. Search Space Geometry

A1.1 Unbounded Search Space:

$$\mathcal{X} = \bigcup_{i=1}^{\infty} \mathcal{E}^i \tag{4}$$

A1.2 Dynamic Search Volume:

$$\forall t \in \mathcal{T}.V(t) = (\mathcal{E}')^{D_{\max}(t)} \tag{5}$$

A1.3 Monotonic Dimension Growth:

$$\forall t \in \mathcal{T}.D_{\max}(t+1) \geq D_{\max}(t) \tag{6}$$

A1.4 Valid Organism (Design Integrity):

$$\forall x \in \mathcal{X}.\text{Valid}(x) \Leftrightarrow (1 \leq \text{len}(x) \wedge \beta(x, \text{len}(x)) = 0 \wedge \forall j.\beta(x, j) \geq 0) \tag{7}$$

A1.5 Closure Under Operations:
The set of valid organisms $\mathcal{X}$ is closed under all genetic operations:

$$\forall x \in \mathcal{X}.\text{Valid}(x) \Rightarrow [\text{Valid}(\text{Mutate}(x)) \wedge \text{Valid}(\text{Crossover}(x, x')) \wedge \text{Valid}(\text{Capture}(x))] \tag{8}$$

This property is maintained by construction through the atomic delimiter pairing constraint.

3.3 A2. Hash Encoding System

A2.1 Core Hash Function:
Let $H : \mathbb{N} \to \mathbb{Z}_{64}$ be defined by:

$$H(i) = \text{xxhash.xxh64_intdigest}(\text{str}(i)) \tag{9}$$

A2.2 Reserved Delimiter Encodings:

$$\varepsilon_{\text{start}} = H(1) \quad \text{(Start delimiter)} \tag{10}$$

$$\varepsilon_{\text{end}} = H(2) \quad \text{(End delimiter)} \tag{11}$$

A2.3 Identifier Management:
Let $c(t)$ denote the gene counter at time t, initialized as $c(0) = 3$.
The system maintains:

- Active encodings: $\mathcal{E}(t) = \{H(i) \mid i \text{ is assigned at time } t\}$
- Available pool: $\mathcal{P}(t) = \{\varepsilon \mid \varepsilon \text{ freed and available for reuse}\}$
- Counter invariant: $|\mathcal{E}(t)| + |\mathcal{P}(t)| \leq c(t) - 3$

A2.4 Gene Encoding Assignment:
For a new gene g:

$$\varepsilon_g = \begin{cases} \text{existing hash key} & \text{if } g \in \text{reverse_encodings} \\ \mathcal{P}(t).\text{pop}() & \text{if } |\mathcal{P}(t)| > 0 \\ H(c(t)),\ c(t+1) \leftarrow c(t) + 1 & \text{otherwise} \end{cases} \tag{12}$$

3.4 A3. Meta gene State Management

A3.1 State Accountability:

$$\forall e, t. \text{IsMetageneEncoding}(e, t) \Leftrightarrow (\text{StateA}(e, t) \oplus \text{StateT}(e, t) \oplus \text{StateP}(e, t)) \tag{13}$$

where $\oplus$ denotes exclusive or (exactly one is true).

A3.2 Information Preservation:

Expansion and inlining operations are bijective by construction:

$$\forall x. I(x) = I(\text{expand}(x)) = I(\text{inline}(x)) \tag{14}$$

This follows from expand(x) replacing encoding e with $\xi(e)$, and inline(x) performing the inverse.

A3.3 LRU Eviction ($\phi_{\mathcal{T}}$):

The Active set $\mathcal{A}(t)$ is a fixed-capacity container with $|\mathcal{A}(t)| \leq \pi_{14}$.

Usage Update Rule:

When Decode(x) processes top-level token e:

$$\text{If IsMetageneEncoding}(e, t): \quad L(e, t) \leftarrow t \tag{15}$$

$$\text{Move } e \text{ to MRU position in } \mathcal{A}(t) \tag{16}$$

Eviction Rule:

When adding new encoding e_{new} to $\mathcal{A}(t)$ and $|\mathcal{A}(t)| = \pi_{14}$:

$$e_{\text{lru}} \leftarrow \arg\min\{L(e, t) \mid e \in \mathcal{A}(t)\} \tag{17}$$

$$\text{StateT}(e_{\text{lru}}, t + 1) \leftarrow \text{true} \tag{18}$$

$$\text{StateA}(e_{\text{lru}}, t + 1) \leftarrow \text{false} \tag{19}$$

$$\text{StateA}(e_{\text{new}}, t + 1) \leftarrow \text{true} \tag{20}$$

Note: Nested meta gene references encountered during recursive expansion do NOT trigger usage updates.

A3.4 Deletion ($\phi_{\mathcal{P}}$):

Age Semantics and Update:

$$\text{Age}_{\mathcal{T}}(e, t) \text{ is defined} \iff \text{StateT}(e, t) \tag{21}$$

$$\forall e, t.\ \text{StateT}(e, t) \ \Rightarrow\ \text{Age}_{\mathcal{T}}(e, t + 1) = \begin{cases} \text{Age}_{\mathcal{T}}(e, t) + 1 & \text{if } \text{Age}_{\mathcal{T}}(e, t) < \pi_{15} \\ \text{Age}_{\mathcal{T}}(e, t) & \text{if } \text{Age}_{\mathcal{T}}(e, t) \geq \pi_{15} \end{cases} \tag{22}$$

Deletion Rule:

$$\text{StateT}(e, t) \wedge \text{Age}_{\mathcal{T}}(e, t) \geq \pi_{15} \ \Rightarrow\ \begin{bmatrix} \forall e' \in \mathcal{A}(t) \cup \mathcal{T}(t).\ e \in \xi(e') \rightarrow \text{inline}(e', e) \\ \text{StateP}(e, t + 1) \wedge \neg\text{StateT}(e, t + 1) \end{bmatrix} \tag{23}$$

where inline(e', e) replaces references to e within e' with $\xi(e)$.

Note: No population membership check. Deletion is purely age-based.

4 Probabilistic Axioms

Let P_{Π} denote the probability measure parameterized by the fixed vector Π.

4.1 P1. Crossover

P1.1 Valid Cut Positions:

$$\text{DepthZeroCuts}(x) = \{j \mid \text{NonDelimitedCut}(x, j)\} \tag{24}$$

P1.2 Crossover Likelihood:

$$P_{\Pi}(\text{Crossover}(p_1, p_2)) = \begin{cases} \pi_{10} & \text{if } |\text{DepthZeroCuts}(p_1)| > 0 \wedge |\text{DepthZeroCuts}(p_2)| > 0 \\ 0 & \text{otherwise} \end{cases} \tag{25}$$

P1.3 Crossover Operation:
If crossover occurs:

$$j_1 \sim \text{Uniform}(\text{DepthZeroCuts}(p_1)) \tag{26}$$
$$j_2 \sim \text{Uniform}(\text{DepthZeroCuts}(p_2)) \tag{27}$$
$$\text{offspring}_1 = p_1[1 : j_1] \oplus p_2[j_2 + 1 : \text{len}(p_2)] \tag{28}$$
$$\text{offspring}_2 = p_2[1 : j_2] \oplus p_1[j_1 + 1 : \text{len}(p_1)] \tag{29}$$

Key Properties:

1. Cuts are INDEPENDENT - positions need not align
2. Each parent cut at depth-zero position (outside all delimiter pairs)
3. Complete delimiter-balanced segments are inherited
4. Both offspring are Valid by construction

4.2 P2. Mutation

P2.1 Context Function:
Define context function $C : \mathbb{N} \times \mathcal{X} \to \{0, 1, 2\}$:

$$C(i, x) = \begin{cases} 0 & \text{if position } i \text{ is undelimited} \\ 1 & \text{if position } i \text{ is delimited} \\ 2 & \text{if } x_i \in \{\text{Start}, \text{End}\} \end{cases} \tag{30}$$

P2.2 Mutation Likelihood (Single Roll per Position):
For position i in organism x, let context $:= C(i, x)$.
If context = 1 (delimited, non-delimiter):

$$\begin{aligned} W = \{&\text{point} : \pi_5/4, \quad \text{swap} : \pi_5/4, \quad \text{insertion} : \pi_5/4, \quad \text{deletion} : \pi_5/4, \\ &\text{capture} : \pi_8, \quad \text{open_no_delimit} : \pi_7, \quad \text{no_op} : 1 - (\pi_5 + \pi_7 + \pi_8)\} \end{aligned} \tag{31}$$

If context = 0 (undelimited):

$$\begin{aligned} W = \{&\text{point} : \pi_4/4, \quad \text{swap} : \pi_4/4, \quad \text{insertion} : \pi_4/4, \quad \text{deletion} : \pi_4/4, \\ &\text{insert_delimiter_pair} : \pi_9, \quad \text{open} : \pi_7 \cdot \nVdash[\text{IsMetageneEncoding}(x[i], t)], \\ &\text{no_op} : 1 - (\pi_4 + \pi_7 + \pi_9)\} \end{aligned} \tag{32}$$

If context = 2 (delimiter token):

$$W = \{\text{delimit_delete} : \pi_6, \quad \text{swap} : (1-\pi_6) \cdot \mathbb{1}[\text{can_swap}(i)], \quad \text{no_op} : \text{residual}\} \tag{33}$$

Selection:

$$\text{roll} \sim \text{Uniform}[0,1] \tag{34}$$

$$\text{mutation_type} := \text{weighted_choice}(W, \text{roll}) \tag{35}$$

4.3 P3. Gene Selection

P3.1 Gene Selection Probabilities:
Let $\mathcal{G}$ be the base gene set and $\mathcal{M}$ be the meta gene stack.

$$P_{\Pi}(G \mid \psi) = \begin{cases} \dfrac{\pi_{12}}{|\mathcal{G}|} & \text{if } G \in \mathcal{G} \\ (1-\pi_{12}) \cdot \dfrac{\pi_{13}^{|\mathcal{M}|-j-1}}{\sum_{k=0}^{|\mathcal{M}|-1} \pi_{13}^{|\mathcal{M}|-k-1}} & \text{if } G = M_j \in \mathcal{M} \end{cases} \tag{36}$$

5 Metagene Operations

5.1 A4. Meta gene Capture

A4.1 Delimiter Pair Location:
find_delimiters$(x, i) \to (s, e)$ where:

$$s = \max\{j < i \mid x[j] = \text{start_delim}\} \tag{37}$$

$$e = \min\{j > s \mid x[j] = \text{end_delim}\} \tag{38}$$

Returns innermost enclosing pair or $\perp$.
A4.2 Content Extraction:

$$\sigma = x[s+1 : e] \quad \text{(Exclude delimiters)} \tag{39}$$

A4.3 Size Constraint:

$$\text{len}(\sigma) \geq 2 \quad \text{(Enforced at call site)} \tag{40}$$

A4.4 Capture Result:
If constraint met:

$$(x', e_{\text{new}}) \quad \text{where } x' \text{ has } \sigma \text{ replaced by } e_{\text{new}} \tag{41}$$

$$\text{StateA}(e_{\text{new}}, t+1) \leftarrow \text{true} \tag{42}$$

$$\xi(e_{\text{new}}) \leftarrow \sigma \tag{43}$$

A4.5 Deduplication:
Before creating new encoding:

$$\text{If } \exists e \in \mathcal{A}(t) \cup \mathcal{T}(t).\xi(e) = \sigma : \text{ Return } e \tag{44}$$

5.2 A5. Meta gene Open

A5.1 Context-Dependent Open Semantics:

For meta gene M at position i with context $C(i, x)$:

Outside delimiters ($C(i, x) = 0$):

$$\text{open}(M) = [\text{Start}] \oplus \xi(M) \oplus [\text{End}] \tag{45}$$

Inside delimited region ($C(i, x) = 1$):

$$\text{open_no_delimit}(M) = \xi(M) \tag{46}$$

Rationale: Opened content outside delimiters receives protection boundaries. Content already inside delimiters inherits existing protection.

A5.2 Delimiter Depth:

Delimiter depth at position j is computed as:

$$\beta(x, j) = \sum_{k=1}^{j-1} (\mathbb{1}[x_k = \text{start_delim}] - \mathbb{1}[x_k = \text{end_delim}]) \tag{47}$$

Note: Sum excludes position j itself. Thus:

- $\beta(x, 1) = 0$ (before first token)
- $\beta(x, j) =$ depth entering position j

A5.3 Functional Invariance:

$$\forall x. F(x) = F(\text{Decode}(x)) \tag{48}$$

An organism's fitness is identical whether evaluated with meta gene references or fully expanded.

6 Population Structure

6.1 A6. Elitism and Special Organisms

A6.1 Elite Preservation:

Let $E(t) = \{x \in P_t \mid \text{Rank}(x, t) \leq \lfloor \pi_{11} \cdot \pi_1 \rfloor\}$

Elite Inheritance:

$$\forall x \in E(t). x \in P_{t+1} \text{ without mutation or crossover} \tag{49}$$

A6.2 Fully Delimited Organism Bypass:

$$\text{IsFullyDelimited}(x) \Leftrightarrow [x[1] = \text{start_delim} \wedge x[\text{len}(x)] = \text{end_delim}] \tag{50}$$

Protection Rule:

$$\text{IsFullyDelimited}(x) \rightarrow [\text{Crossover}(x, \cdot) = (x, \cdot) \wedge \text{Mutate}(x) = x] \tag{51}$$

7 Edge Cases and Fallback Behaviors

7.1 A7. Edge Case Specifications

A7.1 Capture No Valid Pair:
If perform_capture cannot find delimiters:

$$\text{Result} : (x, i+1, \perp) \quad \text{(Advance index, no change)} \tag{52}$$

A7.2 Open Non-Meta gene:
If open_metagene(e) where $\neg$IsMetageneEncoding(e, t):

$$\text{Result} : [e] \quad \text{(Return singleton)} \tag{53}$$

A7.3 Swap Delimiter-Delimiter Forbidden:

$$\text{can_swap}(i, j) \text{ requires: } (x[i] \notin \{\text{start_delim}, \text{end_delim}\}) \vee (x[j] \notin \{\text{start_delim}, \text{end_delim}\}) \tag{54}$$

8 Initialization Process

8.1 A8. Population Initialization

A8.1 Organism Length Sampling:

$$\forall x \in P_0.\text{len}(x) \sim \text{Uniform}\{2, \ldots, \pi_3\} \tag{55}$$

A8.2 Gene Sampling:

$$\forall i \in \{1, \ldots, \text{len}(x)\}.x[i] \sim \text{Uniform}(\mathcal{G}) \tag{56}$$

A8.3 Delimiter Injection (Optional):
If delimiters enabled:

- For each spacing interval of size s
- With probability 0.10: Insert [Start, ..., End] pair
- Constraint: minimum spacing s between pairs

9 System Evolution

9.1 A9. Total System Evolution (Ψ)

A9.1 State Transition:

$$P_\Pi(S(t+1) \mid S(t)) = P_\Pi[\Omega_3(\Omega_2(\Omega_1(S(t))))] \tag{57}$$

A9.2 Operator Sequence (Generational Order):

1. Ω_1 Generation Start: $G(t+1) = G(t) + 1$
2. Ω_2 Generative Phase (Φ): $P_{t+1} = \Phi(P_t, \Pi)$

3. Ω_3 Homeostasis Phase:
   (a) Update $\text{Age}_{\mathcal{T}}$
   (b) Apply deletion
   (c) Apply eviction (usage-triggered during decoding)

A9.3 Detailed Per-Generation Sequence:

1. Increment generation counter
2. Evaluate fitness for P_t
   $\Rightarrow$ Triggers decode operations
   $\Rightarrow$ Updates LRU during evaluation
3. Select parents based on fitness
4. Apply elitism
5. Generate offspring via crossover/mutation
6. Age deletion basket
7. Delete aged meta genes

Note: LRU updates occur during step 2 (fitness evaluation), before selection. Meta gene usage patterns from the current generation influence eviction decisions.

A9.4 Generative Likelihood:

Define:

- $N_{\text{elite}} = \lfloor \pi_{11} \cdot \pi_1 \rfloor$: Number of preserved elite organisms
- $N_{\text{offspring}} = \pi_1 - N_{\text{elite}}$: Number of offspring to generate
- $S(P_t)$: Deterministic parent set (top π_2 organisms ranked by fitness)
- $\text{Pair}(i, S)$: Deterministic pairing function returning parent pair for i-th offspring via round-robin with shift

The population transition factors $P_t \to P_{t+1}$ as $P_{t+1} = E(t) \cup O_{t+1}$ where the preservation of the elite is deterministic ($P(E(t) \mid P_t) = 1$) and the generation of offspring is stochastic.

Revised Generative Likelihood:

$$\log P_{\Pi}(\Phi) = \sum_{i=1}^{N_{\text{offspring}}} \log P_{\Pi}(o_i \mid \text{Pair}(i, S(P_t)), \Pi) \tag{58}$$

where $P_{\Pi}(o_i \mid \text{parents}, \Pi)$ is the probability that the offspring o_i resulting from the combined Crossover (P1) and Mutation (P2) processes applied to its deterministically assigned parent pair.

Key Properties:

1. Preservation of elites and parent selection are deterministic given the fitness ranking
2. Parent pairing follows deterministic round-robin assignment
3. Stochasticity arises only from crossover/mutation operations
4. Offspring generation events are conditionally independent given their parent pairs

10 Geometric Interpretation

10.1 A10. Hyperbolic Model

A10.1 Hyperboloid Model:

Let $\mathbb{H}^n = \{x \in \mathbb{R}^{n+1} : \langle x, x\rangle_M = -1, x_0 > 0\}$ be the hyperboloid model with Minkowski inner product:

$$\langle x, y\rangle_M = -x_0 y_0 + \sum_{j=1}^{n} x_j y_j \tag{59}$$

A10.2 Embedding and Return:

Embedding $\phi : \mathcal{X} \to \mathbb{H}^n$:

$$\phi(x) = \left(\sqrt{1 + \|x\|^2}, x_1, \ldots, x_n\right) \tag{60}$$

Return map $\psi : \mathbb{H}^n \to \mathbb{R}^n$:

$$\psi(x_0, \ldots, x_n) = (x_1, \ldots, x_n) \tag{61}$$

A10.3 Distance Metrics:

Euclidean distance: $d_E(x, y) = \|x - y\|$

Hyperbolic distance: $d_H(x', y') = \operatorname{arccosh}(-\langle x', y'\rangle_M)$ for $x', y' \in \mathbb{H}^n$

11 Conservation Laws

11.1 A11. Invariants

A11.1 Information Conservation:

Total information is preserved by reversible meta gene expansions:

$$I(x) = I(R(x)) \tag{62}$$

for any representation $R(x)$ of sequence x.

A11.2 Structural Integrity:

Define the balance function $\beta(x) = \sum_i (\nVdash_{\text{Start}}(x_i) - \nVdash_{\text{End}}(x_i))$.

Valid sequences satisfy:

1. $\beta(x) = 0$ (balanced delimiters)
2. $1 \leq \text{len}(x) \leq \pi_3$ (length constraints for $t = 0$ only)
3. All prefixes have $\beta(\text{prefix}) \geq 0$ (proper nesting)

A11.3 Identifier Accounting:

$$|\mathcal{A}(t)| + |\mathcal{T}(t)| + |\mathcal{P}(t)| = |\mathcal{E}(t)| \tag{63}$$

Fresh capture occurs when $\mathcal{P}(t) = \emptyset$ increments the total by 1; otherwise, reuse maintains a constant count.

12 Summary

MEGA unifies evolutionary search with dynamic pattern formation through:

- Base space $\mathcal{X}$ with hash encoding $H : \mathbb{N} \to \mathbb{Z}_{64}$
- Emergent meta genes with triple LRU state $(\mathcal{A}, \mathcal{T}, \mathcal{P})$
- $\theta = 2$ deletion threshold
- Context-dependent mutation operators with delimiter awareness
- Independent-cut depth-zero crossover
- Information-preserving evolution maintaining structural invariants

The framework bridges implementation mechanics (hash encoding, LRU cache, operator weights) with analytical tools (hyperbolic geometry, network theory, compression metrics) to provide both algorithmic specification and theoretical understanding of emergent pattern dynamics.